Reinforcement Learning for Adaptive Crowd Flow Control Using Dynamic Barriers

Alhanof Alolyan[1]*, He Wang[2], and Amani Alonayzan[3]

[1] Computer Science, Princess Nourah University
[2] Computer Science, University College London
[3] Independent Researcher
* Corresponding author: hano.alolyan@gmail.com

Publication note. An earlier version of this work was presented at the 26th Scientific Forum for Hajj, Umrah and Visit Research, organised by Umm Al-Qura University, Saudi Arabia, in 2026. This preprint is an English-language author manuscript with minor editorial revisions.

Abstract

Crowd barriers are widely used to regulate pedestrian movement during large public events such as pilgrimages. However, fixed barrier layouts often create unnecessary stopping and local congestion even when they successfully control the overall crowd flow. Finding better barrier configurations is difficult because there is no dataset that provides the best barrier placement for a given crowd scene. We address this challenge by introducing a simulation-driven reinforcement learning framework that learns barrier configurations without requiring labelled training data. Starting from a segmented crowd scene, we construct a realistic simulation environment that reproduces the observed pedestrian distribution. This environment is then used to train a Soft Actor-Critic (SAC) agent that continuously adjusts barrier positions to improve pedestrian movement. Unlike supervised approaches the proposed framework allows the agent to explore barrier configurations that have never been observed in real data. We evaluate the learned policy under unseen numbers of pedestrians. The learned policy consistently outperforms the existing manually designed barrier configuration by reducing pedestrian stops while maintaining effective flow regulation. These results demonstrate that combining computer vision crowd simulation and reinforcement learning provides an effective framework for optimizing crowd control infrastructure. The source code is available on GitHub: https://github.com/HanoCat/RLForHajjProject.git

Keywords: Crowd simulation, reinforcement learning, Soft Actor-Critic (SAC), barrier optimization, pedestrian flow, crowd management, autonomous decision making.

Introduction

Crowd barriers are commonly used to regulate pedestrian movement in dense public spaces. However, manually designed layouts can create unnecessary stopping behaviour and local congestion. This is especially problematic when barriers are intended to regulate the overall crowd flow while avoiding inefficient movement locally. Existing crowd simulation methods mainly focus on modelling pedestrian behaviour or providing interactive tools for manually designing crowd control layouts rather than learning barrier configurations automatically [1][2]. Reinforcement learning has also been applied to crowd simulation and evacuation planning by learning pedestrian behaviours or tuning simulation parameters [3] [4]. However, learning how to optimize physical barrier configurations from an observed crowd scene has less attention.

We propose a simulation-driven framework for adaptive crowd barrier control. Starting from a segmented crowd image, we construct a pedestrian simulation that reproduces the observed crowd distribution. This simulation serves as the environment for (SAC) that learns continuous barrier configurations. The reward encourages smooth pedestrian movement by reducing stopping behaviour and local congestion while preserving the intended flow regulation of the barriers. Unlike supervised learning, our framework does not require labelled examples of optimal barrier placements and allows exploration beyond manually designed layouts. Experiments demonstrate that the learned policy consistently outperforms the existing ground-truth configuration and generalizes to unseen crowd sizes.

Methodology

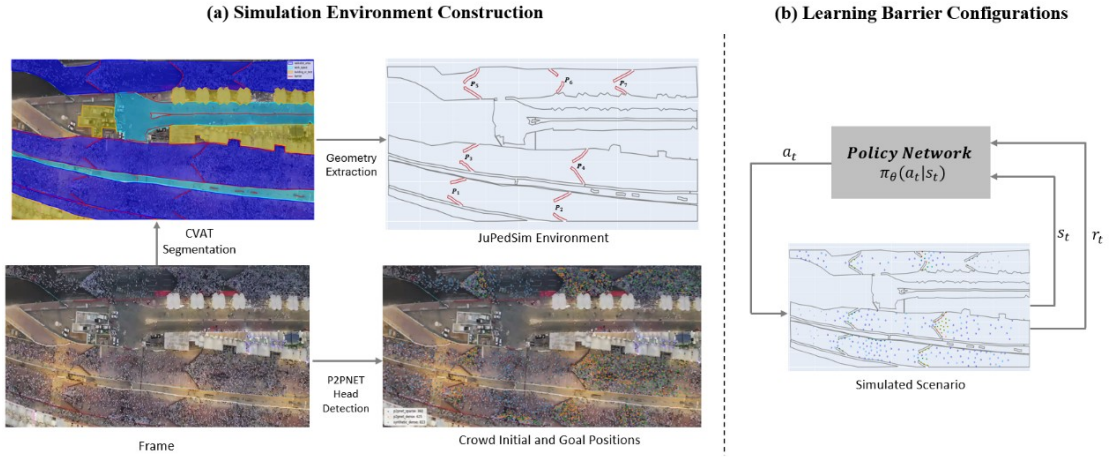


Figure 1. Proposed framework for learning adaptive barrier configurations in pilgrimage crowd management. (a) Semantic CVAT segmentation and P2PNet head detection are used to transform real-world crowd scenes into a JuPedSim environment with extracted geometry and crowd initialization states. (b) The reinforcement learning policy learns adaptive barrier configurations through interaction with the simulated crowd dynamics, avoiding manually engineered control rules and threshold selection.

Figure 1 illustrates the proposed framework, which consists of two main stages. First, a crowd image, we first annotate the walkable regions, obstacles, and barrier locations using CVAT [5]. The extracted geometry is converted into a JuPedSim simulation environment [6]. Pedestrian head locations are detected using P2PNet [7] and each pedestrian is assigned an initial position and a destination according to the segmented scene. This process produces a realistic simulation that preserves the observed crowd distribution while allowing different barrier configurations to be evaluated. Unlike collecting real barrier-control data, the simulation provides unlimited interactions for reinforcement learning.

Second, the generated simulation is formulated as a reinforcement learning environment where (SAC) learns continuous barrier configurations. At each time step t, the state is defined as $s_t = \{N, B_t, P_t\}$, where N denotes the number of pedestrians, B_t represents the current configuration of all barrier pairs, and $P_t = \{(x_i, y_i, v_i)\}_{i=1}^{N}$ denotes the simulated crowd state, including the position and speed of each pedestrian. The action vector a_t continuously adjusts the translation $(\Delta x, \Delta y)$ and rotation $(\Delta\theta)$ of every barrier pair. The reward is defined as $r_t = -C$, where

$$C = \lambda_s R_{stop} + \lambda_v \left(1 - \frac{\acute{v}}{v_{max}}\right),$$

where R_{stop} is the proportion of stopped pedestrians, $\acute{v}$ is the mean pedestrian speed, v_{max} is the desired maximum walking speed, and $\lambda_s + \lambda_v = 1$. This objective encourages barrier configurations that reduce stopping while maintaining smooth pedestrian movement.

Results and Discussion

Figure 2 shows the training process of the proposed SAC agent. The critic losses decrease steadily while the actor converges to a stable policy, indicating consistent learning throughout training. Figure 3 compares the learned policy with the ground-truth (GT), all-open, and all-closed barrier configurations across different crowd sizes, including unseen numbers of pedestrians.

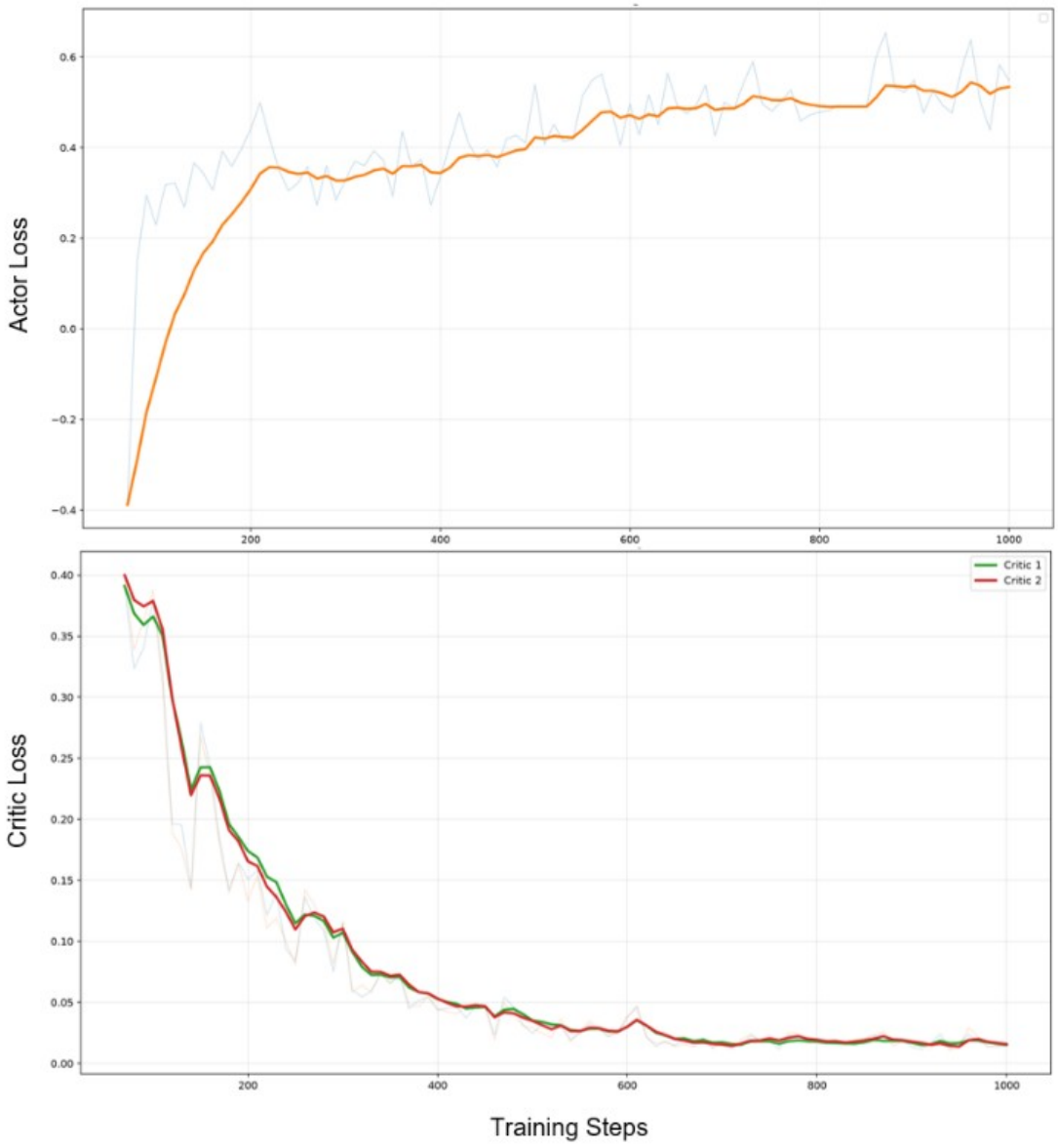


Figure 2 . Training curves of the proposed RL agent over training steps. The top plot shows the actor loss, while the bottom plot presents the losses of the two critic networks. Faint lines denote the raw training values, and solid lines show the exponentially smoothed trends. The critic losses steadily decrease and converge, indicating stable value-function learning, while the actor loss gradually stabilizes as the policy converges.

The all-open configuration achieves the highest throughput and lowest stopping ratio because pedestrians move without physical restrictions. However, completely removing barriers is not a practical solution since barriers are required to regulate pedestrian flow at the global level. In contrast, the all-closed configuration produces severe congestion and the poorest performance.

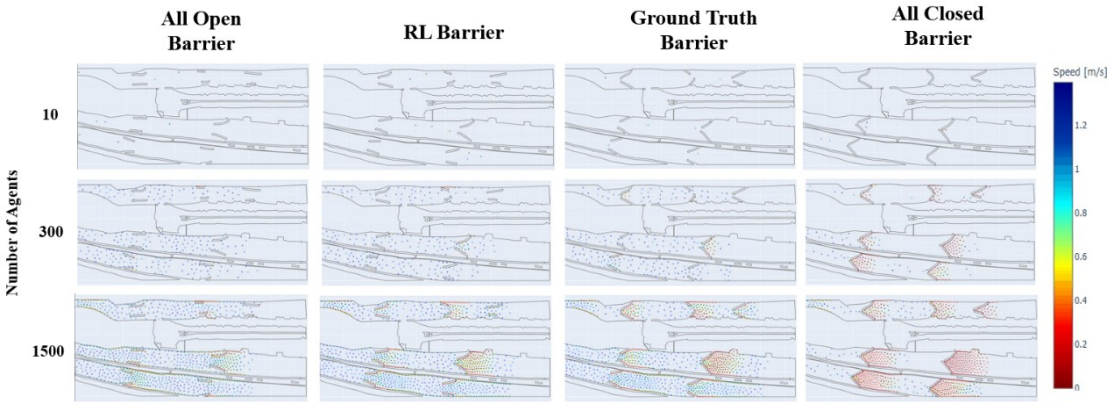


Figure 3 Qualitative comparison of barrier configurations under different crowd sizes. The proposed RL policy reduces local congestion while preserving the intended flow regulation compared with the existing ground-truth (GT) configuration. Results are shown for 10, 300, and 1500 pedestrians. The policy was trained only on crowd sizes between 100 and 700 pedestrians. Therefore, the scenarios with 10 and 1500 pedestrians demonstrate the policy's ability to generalize to unseen number of agents.

The proposed policy learns a better compromise by adapting barrier configurations to the current crowd density. Compared with the GT layout, our method increases throughput (5.098 vs. 4.604 agents/s), improves the normalized pedestrian velocity (0.942 vs. 0.887), and reduces the stopped ratio (0.072 vs. 0.104).

Table 1. Overall performance comparison across all evaluation scenarios. The table reports the mean and standard deviation of pedestrian throughput, normalized velocity, and stopped ratio for the proposed RL policy, the random baseline, and the reference barrier configurations. Arrows (↑/↓) indicate whether higher or lower values are desirable.

Configuration	Stopped ratio ↓	Normalized velocity ↑	Throughput (agents/s) ↑
GT	0.104 ± 0.076	0.887 ± 0.223	4.604 ± 2.966
All Open	0.035 ± 0.023	1.000 ± 0.134	5.447 ± 3.544
All Closed	0.642 ± 0.075	0.304 ± 0.097	1.991 ± 1.287
RL (ours)	0.072 ± 0.075	0.942 ± 0.155	5.098 ± 3.302

Figure 4 further shows that the learned configurations consistently reduce local congestion while maintaining effective crowd regulation across different number of agents.

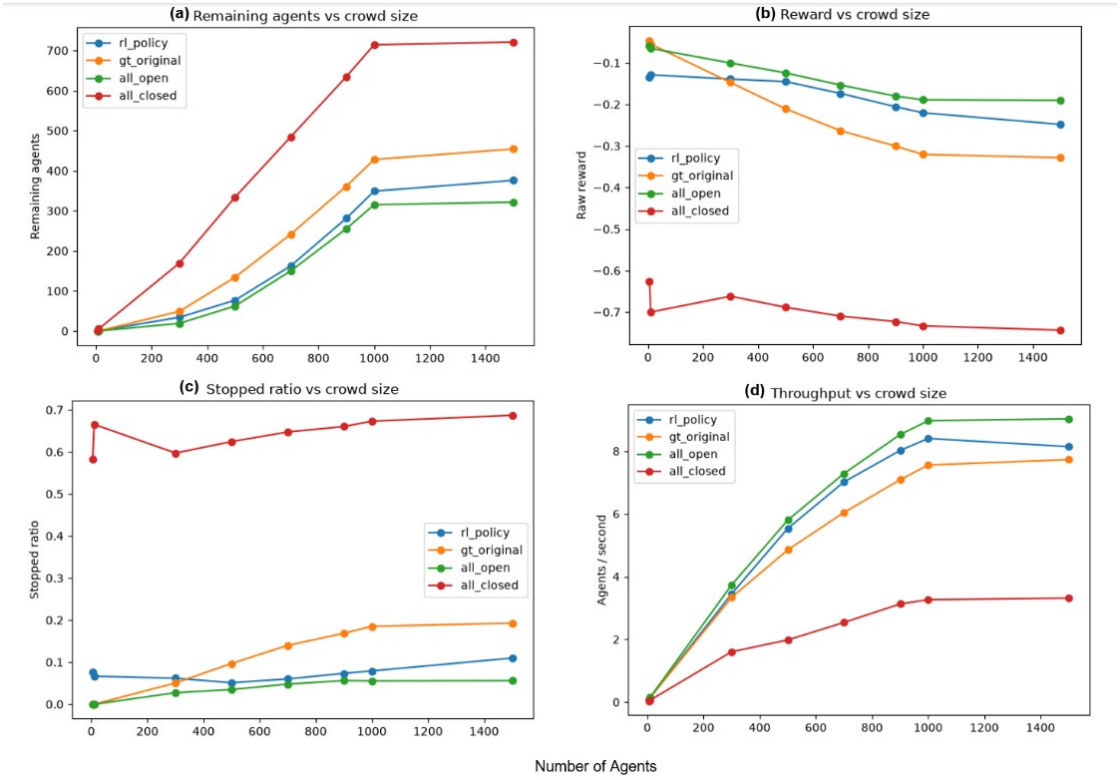


Figure 4. Performance comparison between the proposed RL-based adaptive barrier control (Ours) and the original barrier configuration (Ground Truth) under increasing crowd sizes. (a) Number of remaining agents at the end of the simulation (lower is better). (b) Mean reward obtained by each method. (c) Ratio of stopped agents (lower is better). (d) Pedestrian throughput (agents/s, higher is better). The proposed method maintains lower congestion and higher throughput as crowd density increases.

Conclusions

This paper presents a practical framework for learning crowd barrier configurations in the absence of optimal training data are unavailable. Instead of relying on manually designed layouts or predefined control rules, the proposed approach constructs a learning environment directly from an observed crowd scene and enables reinforcement learning to discover effective barrier configurations through interaction. Compared with the existing ground-truth (GT) barrier configuration, the proposed method increased pedestrian throughput by approximately **10.73%** (from 4.604 to 5.098 agents/s), improved normalized pedestrian velocity by approximately **6.20%** (from 0.887 to 0.942), and reduced the stopped-pedestrian ratio by approximately **30.77%** (from 0.104 to 0.072). These results demonstrate that the learned policy provides a better balance between improving pedestrian movement and preserving the intended role of barriers in regulating crowd flow. Furthermore, the policy generalizes to unseen crowd sizes without additional supervision or scene-specific tuning, demonstrating the potential of simulation-driven reinforcement learning for intelligent crowd-management systems.

Recommendations

- Incorporate perspective correction and scene calibration using known measurements, such as corridor widths, barrier dimensions, or site maps, to establish an accurate pixel-to-meter transformation.
- Investigate faster simulation engines to reduce the computational cost associated with the large number of pedestrian simulations required during reinforcement learning.

- Explore more sample-efficient reinforcement learning methods to reduce training requirements and improve the scalability of the proposed framework.
- Extend the proposed framework to multiple crowd scenes and live crowd observations to evaluate its generalizability and support future real-world deployment.

References

[1] Montana Gonzalez, L.R. and Maddock, S., 2021. A sketch-based interface for real-time control of crowd simulations that incorporate dynamic knowledge. Journal of Virtual Reality and Broadcasting, 16(3).

[2] Charalambous, P., Pettre, J., Vassiliades, V., Chrysanthou, Y. and Pelechano, N., 2023. Greil-crowds: Crowd simulation with deep reinforcement learning and examples. *ACM Transactions on Graphics (TOG)*, *42*(4), pp.1-15.

[3] Zhao, Y. and Geraerts, R., 2022, August. Automatic Parameter tuning via reinforcement learning for crowd simulation with social distancing. In 2022 26th International Conference on Methods and Models in Automation and Robotics (MMAR) (pp. 87-92). IEEE.

[4] Zhang, Y., Yang, B. and Zhu, J., 2024. A double-layer crowd evacuation simulation method based on deep reinforcement learning. *Computer Animation and Virtual Worlds*, *35*(3), p.e2280.

[5] Sekachev, B., Manovich, N. and Zhavoronkov, A., 2019. Computer vision annotation tool. *Zenodo*.

[6] Wagoum, A.K., Chraibi, M., Zhang, J. and Lämmel, G., 2015, December. JuPedSim: an open framework for simulating and analyzing the dynamics of pedestrians. In *3rd Conference of Transportation Research Group of India* (Vol. 12).

[7] Song, Q., Wang, C., Jiang, Z., Wang, Y., Tai, Y., Wang, C., Li, J., Huang, F. and Wu, Y., 2021. Rethinking counting and localization in crowds: A purely point-based framework. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3365-3374).